
AVR Toolchain Installer: Release 3.2.3.579

The AVR Toolchain contains software development toolchains for both the 8-bit and 32-bit AVR device families. In the packages is GNU Toolchains for 8-bit and 32-bit AVR based on the free and open-source GCC compiler. Though it may not be permanently tied to only this compiler in the future.

The AVR Toolchain inherits from two previous software distributions, the AVR32 GNU Toolchain, and WinAVR. The AVR32 GNU Toolchain was a software development toolchain for the AVR 32-bit devices only, and was developed within Atmel. WinAVR was a software development toolchain for the AVR 8-bit devices only, and was developed outside of Atmel.



8/32-bit AVR[®]
Microcontrollers

Release 3.2.3.579





Installation Instructions

AVR Toolchain is a suite of executable, open source software development tools for the Atmel AVR 8/32-bit series of Micro Controller Units (MCU). It includes the GNU GCC compiler for C and C++.

Installation

The package comes in several forms.

- As a standalone installer
- As part of AVR Studio 5

Mainfest

1. AVR 8-bit GNU Toolchain
 - Compiler (GCC) , Binary utilities (including assembler, linker, etc.), debugger (GDB) and C-library (avr-libc) for AVR 8 bits devices.
2. AVR 32-bit GNU Toolchain
 - Compiler (GCC) , Binary utilities (including assembler, linker, etc.), debugger (GDB) and C-library (Newlib) for AVR 32 bits devices.
3. AVR 8 Utilities
 - Utilities for programming AVR 8-bit devices
4. Documentation for the various projects.
5. Source code patches used to build the various projects.

Layout

Listed below are some folders you might want to know about.

`<install>` = The directory where you installed AVR Toolchain.

- <install>
 - The folder where you installed AVR Toolchain.
 - Default folder when using the installer is 'C:\Program Files\Atmel\AVR Tools\AVR Toolchain'
- <install>\bin
 - The folder where all executables are installed
- <install>\source
 - The folder where patches for different open source programs is located
- <install>\doc
 - The folder where Readme files and other documentations are installed

For more details see Release Notes for AVR 8-bit GNU Toolchain and AVR 32-bit GNU Toolchain.

`PATH` Environment Variable

There is one folder in AVR Toolchain that contain executable programs. If `<install>` is your install folder then this folder is:

<install>\bin

AVR Toolchain Installer

The ``<install>\bin`` folder contains the software development toolset proper. This includes GNU binutils, GCC, and other programs and many miscellaneous Unix or GNU programs that are built for Windows. This includes sh (bash) and make among a host of other things.

For your operating system to easily locate these folders, they must be put at the **beginning** of the ``PATH`` environment variable. The reason for putting these directories at the beginning of the ``PATH`` environment variable is for the correct make program to be called.

AVR Toolchain installer modifies this registry key: ``HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Session Manager\Environment\Path``

IMPORTANT: On Windows NT/2K/XP you must have Administrator privileges for the installer to automatically put these directories in your ``PATH`` environment variable.

Registry Keys

AVR Toolchain installs a minimal amount of registry keys. These keys are installed to inform of the installation path, the uninstaller, and for GCC to find other parts of the compiler as needed. Below are the specific keys that are installed.

- **Installation Location**

- This registry key will be added to provide the location of the AVR Toolchain installation:

"`HKEY_LOCAL_MACHINE\SOFTWARE\Atmel\AVR Toolchain\{VERSION}`"

with {VERSION} being replaced by the version number of AVR Toolchain.

Formerly, the key above without {VERSION} was equal to the install location.

- **GCC Component Paths**

- There are some keys that are installed that are used to help GCC find installed component programs:

"`HKEY_LOCAL_MACHINE\Software\Free Software Foundation\AVR Toolchain\{VERSION}\GCC`"

"`HKEY_LOCAL_MACHINE\Software\Free Software Foundation\AVR Toolchain\{VERSION}\BINUTILS`"

"`HKEY_LOCAL_MACHINE\Software\Free Software Foundation\AVR Toolchain\{VERSION}\G++`"

with {VERSION} being replaced by the version number of AVR Toolchain. Each of these keys points to the AVR Toolchain install location for that version of AVR Toolchain.

- **PATH Environment Variable**

- The registry key:

"`HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Session Manager\Environment\Path`"

may be modified (if selected at installation time) to add two directories to the PATH environment variable.

- **Uninstaller**

- There are several registry keys written under:

"`HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Uninstall\{Product GUID}`"

with Product-GUID being 2CE5E313-EC49-4527-A752-6DC89FE51C0D. These registry keys are used to register the uninstaller with Windows.

Finding Help

Documentation

The first and best place to find help is in the documentation! AVR Toolchain includes the user manuals for many of the software tools that are shipped in the package.





The documentation for any particular package may come in different formats depending upon what is available from that package and available space in the AVR Toolchain installation. The different documentation formats that you'll find in AVR Toolchain are:

1. HTML - Hyper Text Markup Language. Requires a web browser to view.
2. PDF - Portable Document Format. Requires a PDF viewer such as Acrobat.

Additionally many user manuals can also be found online, especially packages that are part of the GNU project. You can find links to many of these in the Links section below.

For packages that have HTML, and PDF documentation, look in your `<install>\doc`` directory.

AVR Toolchain installs on your desktop two shortcuts. One is to the HTML documentation on avr-libc that is installed locally. The other shortcut is to the GNU Manuals online (which requires Internet connection).

Online Sources

Please try and find the answer in the documentation first before asking for help online.

TIP: If you need to ask for help online, please read this first: How To Ask Questions The Smart Way (<http://catb.org/~esr/faqs/smart-questions.html>)

Help for the AVR software development toolset (and specifically AVR GCC) can be found at:

- **Atmel AVR Home** (<http://www.atmel.com/avr>)
 - Find info about different Atmel AVR products and application notes.
- **AVR Freaks** (<http://www.avrfreaks.net>)
 - All AVR, all the time! This site has several forums including a general AVR Forum and an AVR GCC Forum specifically for discussion of the GCC compiler for the AVR. They also have an Academy which contains user's projects. This gives you access to a lot of sample code, libraries, and various AVR projects.
- **avr-gcc mailing list** (<http://savannah.nongnu.org/mail/?group=avr>)
 - The avr-gcc list is intended as a forum for discussion about the following: Bugs, Programming technique, Installation and distributions, Hints and tips, Other avr-gcc related stuff. Note that all of the developers of the toolset are subscribed to this list!
- **Mikrocontroller.net** (<http://www.mikrocontroller.net/>)
 - For native German speakers. They have a forum for the AVR GCC compiler.

Help for other projects and tools included in AVR Toolchain can usually be found at the individual project's web page which usually includes links to their mailing lists.

If you need help, do not contact the AVR Toolchain developers personally! Use these web sites and mailing lists first!

Support and Bugs

You can fill out a relevant tracker on the <http://support.atmel.no>, if you have one of the following:

- a bug in the packaging
- a bug in the installation
- a bug in the software

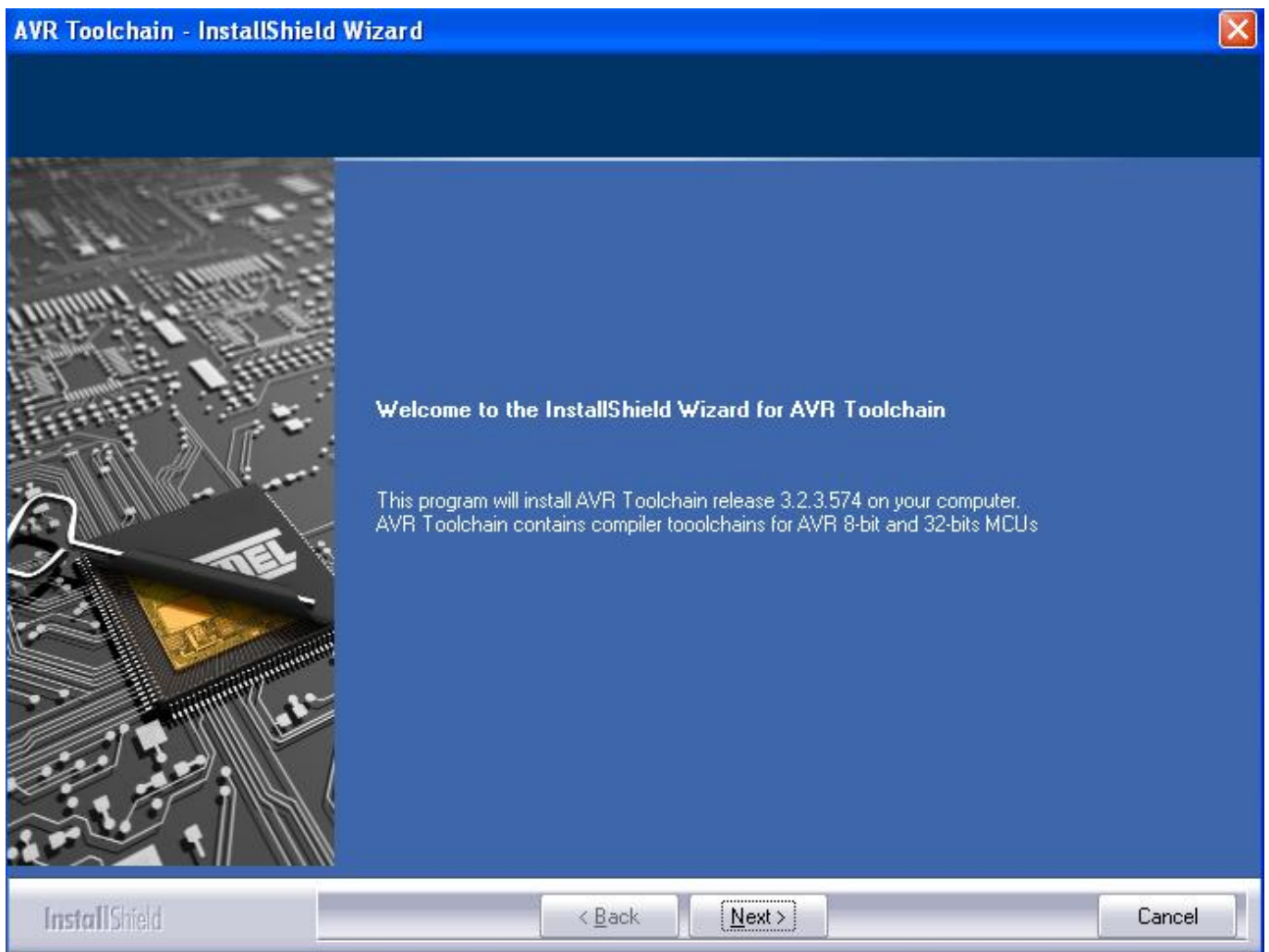
- any suggestions for a new tool to be added
- any suggestions for improvements to the overall package

New and Noteworthy

This chapter lists new and noteworthy items for the AVR Toolchain release.

AVR Toolchain

Replaces WinAVR and AVR32 GNU Toolchain



The *AVR Toolchain* replaces WinAVR and AVR32 GNU Toolchain packages.



Contact Information

For support on AVR Toolchain please contact avr@atmel.com.

Users of AVR Toolchain are also welcome to discuss on the AVRfreaks website forum for AVR Software Tools.

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